

# Master Index

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Portfolio-wide topic, package, endpoint, node, feature, and file index.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

**Source of truth.** Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

## Volume directory

| Vol | Document                                                                                                               | Scope                                                                          |
|-----|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 00  | <b>Codebase Start Here</b><br><a href="#">00_CODEBASE_START_HERE.pdf</a>                                               | Orientation, reading path, setup, and the system mental model.                 |
| 01  | <b>System Architecture</b><br><a href="#">01_SYSTEM_ARCHITECTURE.pdf</a>                                               | Boundaries, services, topology, and principal request flows.                   |
| 02  | <b>Tech Stack and Decisions</b><br><a href="#">02_TECH_STACK_AND_DECISIONS.pdf</a>                                     | Technology inventory and evidence-backed design choices.                       |
| 03  | <b>Backend Code Reference</b><br><a href="#">03_BACKEND_CODE_REFERENCE.pdf</a>                                         | Package, module, class, and function reference for app/.                       |
| 04  | <b>Frontend Code Reference</b><br><a href="#">04_FRONTEND_CODE_REFERENCE.pdf</a>                                       | React modes, components, hooks, API clients, and types.                        |
| 05  | <b>Frontend-Backend Mapping</b><br><a href="#">05_FRONTEND_BACKEND_MAPPING.pdf</a>                                     | UI surfaces mapped to API routes, stores, and runtime behavior.                |
| 06  | <b>Workflow Engine and Node Types</b><br><a href="#">06_WORKFLOW_ENGINE_AND_NODE_TYPES.pdf</a>                         | Workflow contracts, compiler, preflight, registry, and node catalog.           |
| 07  | <b>Runtime Execution and State</b><br><a href="#">07_RUNTIME_EXECUTION_AND_STATE.pdf</a>                               | Execution lifecycle, state transitions, retries, HITL, and resume.             |
| 08  | <b>Data Storage and State Management</b><br><a href="#">08_DATA_STORAGE_AND_STATE_MANAGEMENT.pdf</a>                   | MongoDB, Redis, MinIO, Weaviate, checkpoints, and browser state.               |
| 09  | <b>AI, LLM, and Model Routing</b><br><a href="#">09_AI_LLM_MODEL_ROUTING.pdf</a>                                       | Providers, model routing, fallback, policy, and cost accounting.               |
| 10  | <b>RAG, Knowledge, and Retrieval</b><br><a href="#">10_RAG_KNOWLEDGE_AND_RETRIEVAL.pdf</a>                             | Ingestion, indexing, secure filtering, hybrid search, and grounded generation. |
| 11  | <b>MCP, Tools, and Integrations</b><br><a href="#">11_MCP_TOOLS_AND_INTEGRATIONS.pdf</a>                               | MCP discovery, business systems, email, Drive, and external actions.           |
| 12  | <b>API Reference and Request Flows</b><br><a href="#">12_API_REFERENCE_AND_REQUEST_FLOWS.pdf</a>                       | HTTP route catalog, authorization, payloads, and sequences.                    |
| 13  | <b>Security, Auth, and Data Protection</b><br><a href="#">13_SECURITY_AUTH_AND_DATA_PROTECTION.pdf</a>                 | Identity, RBAC, ownership, isolation, middleware, and sensitive data.          |
| 14  | <b>Events, Async, and Distributed Coordination</b><br><a href="#">14_EVENTS_ASYNC_AND_DISTRIBUTED_COORDINATION.pdf</a> | SSE, Redis streams, background work, leases, and multi-worker behavior.        |
| 15  | <b>Testing, Debugging, and Observability</b><br><a href="#">15_TESTING_DEBUGGING_AND_OBSERVABILITY.pdf</a>             | Test layers, diagnostics, logging, metrics, traces, and QA evidence.           |
| 16  | <b>Deployment, Configuration, and Operations</b><br><a href="#">16_DEPLOYMENT_CONFIGURATION_AND_OPERATIONS.pdf</a>     | Configuration, containers, CI/CD, deployment, backup, and recovery.            |
| 17  | <b>File-by-File Code Reference</b><br><a href="#">17_FILE_BY_FILE_CODE_REFERENCE.pdf</a>                               | Purpose and major symbols for every first-party source file.                   |

| Vol | Document                                                                                               | Scope                                                                            |
|-----|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 18  | <b>Code Logic Reference</b><br><a href="#">18_CODE_LOGIC_REFERENCE.pdf</a>                             | Important functions, methods, branching logic, invariants, and failure behavior. |
| 19  | <b>Docstrings and Code Documentation</b><br><a href="#">19_DOCSTRINGS_AND_CODE_DOCUMENTATION.pdf</a>   | Documentation coverage, public symbols, conventions, and gaps.                   |
| 20  | <b>Architecture Tradeoffs and Tech Debt</b><br><a href="#">20_ARCHITECTURE_TRADEOFFS_TECH_DEBT.pdf</a> | Confirmed compromises, coupling, risks, removals, and remediation guidance.      |
| 21  | <b>Change Impact Guide</b><br><a href="#">21_CHANGE_IMPACT_GUIDE.pdf</a>                               | Change scenarios, affected layers, required tests, and rollout checks.           |
| 22  | <b>Feature-to-Code Map</b><br><a href="#">22_FEATURE_TO_CODE_MAP.pdf</a>                               | Product capabilities mapped to UI, API, runtime, storage, and tests.             |
| 23  | <b>Master Index</b><br><a href="#">23_MASTER_INDEX.pdf</a>                                             | Portfolio-wide topic, package, endpoint, node, feature, and file index.          |

## Master topic and symbol index

| Entry              | Kind                    | Primary source                       | Volumes       |
|--------------------|-------------------------|--------------------------------------|---------------|
| .env.example       | Package/area (1 files)  | <a href="#">.env.example</a>         | 01, 03/04, 17 |
| .github/workflows  | Package/area (3 files)  | <a href="#">.github/workflows</a>    | 01, 03/04, 17 |
| AITaskAgent        | Node type               | <a href="#">app/nodes/ai_task.py</a> | 06, 17, 22    |
| app/__init__.py    | Package/area (1 files)  | <a href="#">app/__init__.py</a>      | 01, 03/04, 17 |
| app/api            | Package/area (29 files) | <a href="#">app/api</a>              | 01, 03/04, 17 |
| app/config.py      | Package/area (1 files)  | <a href="#">app/config.py</a>        | 01, 03/04, 17 |
| app/db             | Package/area (3 files)  | <a href="#">app/db</a>               | 01, 03/04, 17 |
| app/evaluation     | Package/area (6 files)  | <a href="#">app/evaluation</a>       | 01, 03/04, 17 |
| app/evidence       | Package/area (6 files)  | <a href="#">app/evidence</a>         | 01, 03/04, 17 |
| app/ingestion      | Package/area (14 files) | <a href="#">app/ingestion</a>        | 01, 03/04, 17 |
| app/integrations   | Package/area (21 files) | <a href="#">app/integrations</a>     | 01, 03/04, 17 |
| app/knowledge      | Package/area (6 files)  | <a href="#">app/knowledge</a>        | 01, 03/04, 17 |
| app/llm            | Package/area (18 files) | <a href="#">app/llm</a>              | 01, 03/04, 17 |
| app/main.py        | Package/area (1 files)  | <a href="#">app/main.py</a>          | 01, 03/04, 17 |
| app/mcp            | Package/area (28 files) | <a href="#">app/mcp</a>              | 01, 03/04, 17 |
| app/nodes          | Package/area (62 files) | <a href="#">app/nodes</a>            | 01, 03/04, 17 |
| app/observability  | Package/area (5 files)  | <a href="#">app/observability</a>    | 01, 03/04, 17 |
| app/proposal_graph | Package/area (9 files)  | <a href="#">app/proposal_graph</a>   | 01, 03/04, 17 |
| app/rag            | Package/area (3 files)  | <a href="#">app/rag</a>              | 01, 03/04, 17 |
| app/research       | Package/area (3 files)  | <a href="#">app/research</a>         | 01, 03/04, 17 |
| app/retrieval      | Package/area (16 files) | <a href="#">app/retrieval</a>        | 01, 03/04, 17 |

| Entry                                                        | Kind                    | Primary source                                                                                                                                                              | Volumes       |
|--------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| app/runtime                                                  | Package/area (21 files) | <a href="#">app/runtime</a>                                                                                                                                                 | 01, 03/04, 17 |
| app/security                                                 | Package/area (17 files) | <a href="#">app/security</a>                                                                                                                                                | 01, 03/04, 17 |
| app/storage                                                  | Package/area (2 files)  | <a href="#">app/storage</a>                                                                                                                                                 | 01, 03/04, 17 |
| app/tools                                                    | Package/area (14 files) | <a href="#">app/tools</a>                                                                                                                                                   | 01, 03/04, 17 |
| app/workflow                                                 | Package/area (16 files) | <a href="#">app/workflow</a>                                                                                                                                                | 01, 03/04, 17 |
| Authentication                                               | Feature                 | <a href="#">ui/src/components/auth</a> ; <a href="#">app/api/auth.py</a> ; security modules                                                                                 | 22            |
| BoundedDeepResearchAgent                                     | Node type               | <a href="#">app/nodes/bounded_deep_research_agent.py</a>                                                                                                                    | 06, 17, 22    |
| Business Chat and managed sources                            | Feature                 | <a href="#">ui/src/modes/studio/business-chat</a> ; <a href="#">app/api/chat_workspace.py</a> ; <a href="#">app/api/run_chat.py</a> ; chat workflow/conversation/run stores | 22            |
| CallCoverageMatrixAgent                                      | Node type               | <a href="#">app/nodes/call_coverage.py</a>                                                                                                                                  | 06, 17, 22    |
| CitationRegistryBuilder                                      | Node type               | <a href="#">app/nodes/citation_registry_builder.py</a>                                                                                                                      | 06, 17, 22    |
| ClaimEvidenceVerifier                                        | Node type               | <a href="#">app/nodes/claim_evidence_verifier.py</a>                                                                                                                        | 06, 17, 22    |
| ConceptAlternativesAgent                                     | Node type               | <a href="#">app/nodes/concept_alternatives.py</a>                                                                                                                           | 06, 17, 22    |
| ConceptFreezeAgent                                           | Node type               | <a href="#">app/nodes/concept_freeze.py</a>                                                                                                                                 | 06, 17, 22    |
| ConsistencyChecker                                           | Node type               | <a href="#">app/nodes/consistency_checker.py</a>                                                                                                                            | 06, 17, 22    |
| Cost management                                              | Feature                 | <a href="#">ui/src/modes/cost</a> ; cost APIs; cost ledger                                                                                                                  | 22            |
| DataTransformAgent                                           | Node type               | <a href="#">app/nodes/data_transform.py</a>                                                                                                                                 | 06, 17, 22    |
| DecisionAgent                                                | Node type               | <a href="#">app/nodes/decision.py</a>                                                                                                                                       | 06, 17, 22    |
| DELETE /api/builder/email/connections/{connection_id}        | API route               | <a href="#">app/api/email_oauth.py</a>                                                                                                                                      | 12            |
| DELETE /api/builder/integrations/connections/{connection_id} | API route               | <a href="#">app/api/integration_oauth.py</a>                                                                                                                                | 12            |
| DELETE /api/chat-workflows/{chat_workflow_id}                | API route               | <a href="#">app/api/chat_workflows.py</a>                                                                                                                                   | 12            |
| DELETE /api/cost-admin/pricing/{model}                       | API route               | <a href="#">app/api/cost_admin.py</a>                                                                                                                                       | 12            |
| DELETE /api/entity-registry                                  | API route               | <a href="#">app/api/entity_registry.py</a>                                                                                                                                  | 12            |
| DELETE /api/prompt-templates/{template_id}                   | API route               | <a href="#">app/api/prompt_templates.py</a>                                                                                                                                 | 12            |
| DELETE /api/runs/mine/{run_id}                               | API route               | <a href="#">app/api/runs.py</a>                                                                                                                                             | 12            |
| DELETE /api/workflows/{name}                                 | API route               | <a href="#">app/api/workflows.py</a>                                                                                                                                        | 12            |
| DELETE /api/workflows/{name}/draft                           | API route               | <a href="#">app/api/workflows.py</a>                                                                                                                                        | 12            |
| deploy/ionos                                                 | Package/area (8 files)  | <a href="#">deploy/ionos</a>                                                                                                                                                | 01, 03/04, 17 |
| docker-compose.production.yml                                | Package/area (1 files)  | <a href="#">docker-compose.production.yml</a>                                                                                                                               | 01, 03/04, 17 |
| docker-compose.yml                                           | Package/area (1 files)  | <a href="#">docker-compose.yml</a>                                                                                                                                          | 01, 03/04, 17 |

| Entry                                                                        | Kind                   | Primary source                                              | Volumes       |
|------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------|---------------|
| Dockerfile                                                                   | Package/area (1 files) | Dockerfile                                                  | 01, 03/04, 17 |
| DOCXProposalRenderer                                                         | Node type              | app/nodes/docx_renderer.py                                  | 06, 17, 22    |
| DynamicFigureAgent                                                           | Node type              | app/nodes/dynamic_figure_agent.py                           | 06, 17, 22    |
| Echo                                                                         | Node type              | app/nodes/_stubs.py                                         | 06, 17, 22    |
| EmailAgent                                                                   | Node type              | app/nodes/email_integration.py                              | 06, 17, 22    |
| EndAgent                                                                     | Node type              | app/nodes/end.py                                            | 06, 17, 22    |
| Evaluation Lab                                                               | Feature                | ui/src/modes/eval; app/api/eval.py; evaluation package      | 22            |
| ExcelTableExtractor                                                          | Node type              | app/nodes/excel_tool.py                                     | 06, 17, 22    |
| ExternalActionAgent                                                          | Node type              | app/nodes/external_action.py                                | 06, 17, 22    |
| FigureEmbedder                                                               | Node type              | app/nodes/figure_embedder.py                                | 06, 17, 22    |
| Files/artifacts                                                              | Feature                | source/output UI; workflow file routes; MinIO and renderers | 22            |
| GET /api/                                                                    | API route              | app/api/workflows.py                                        | 12            |
| GET /api/audit/session/{session_id}                                          | API route              | app/api/audit.py                                            | 12            |
| GET /api/builder/email/connect/{provider}                                    | API route              | app/api/email_oauth.py                                      | 12            |
| GET /api/builder/email/connections                                           | API route              | app/api/builder.py                                          | 12            |
| GET /api/builder/email/oauth/callback/{provider}                             | API route              | app/api/email_oauth.py                                      | 12            |
| GET /api/builder/integrations/connect/{provider}                             | API route              | app/api/integration_oauth.py                                | 12            |
| GET /api/builder/integrations/connections                                    | API route              | app/api/integration_oauth.py                                | 12            |
| GET /api/builder/integrations/connections/{connection_id}/download/{file_id} | API route              | app/api/integration_oauth.py                                | 12            |
| GET /api/builder/integrations/connections/{connection_id}/files              | API route              | app/api/integration_oauth.py                                | 12            |
| GET /api/builder/integrations/connections/{connection_id}/path/{file_id}     | API route              | app/api/integration_oauth.py                                | 12            |
| GET /api/builder/integrations/oauth/callback/{provider}                      | API route              | app/api/integration_oauth.py                                | 12            |
| GET /api/builder/mcp/servers                                                 | API route              | app/api/builder.py                                          | 12            |
| GET /api/builder/mcp/servers/{server_id}/health                              | API route              | app/api/builder.py                                          | 12            |
| GET /api/builder/mcp/servers/{server_id}/tools                               | API route              | app/api/builder.py                                          | 12            |
| GET /api/builder/operators                                                   | API route              | app/api/builder.py                                          | 12            |
| GET /api/candidates/{run_id}                                                 | API route              | app/api/candidates.py                                       | 12            |

| Entry                                                         | Kind      | Primary source                | Volumes |
|---------------------------------------------------------------|-----------|-------------------------------|---------|
| GET /api/candidates/{run_id}/documents/{document_id}/download | API route | app/api/candidates.py         | 12      |
| GET /api/chat-conversations/{conversation_id}                 | API route | app/api/chat_conversations.py | 12      |
| GET /api/chat-workflows                                       | API route | app/api/chat_workflows.py     | 12      |
| GET /api/chat-workflows/adapters/by-name/{workflow_name}      | API route | app/api/chat_workflows.py     | 12      |
| GET /api/chat-workflows/{chat_workflow_id}                    | API route | app/api/chat_workflows.py     | 12      |
| GET /api/chat-workspace/experiences                           | API route | app/api/chat_workspace.py     | 12      |
| GET /api/cost-admin/budgets                                   | API route | app/api/cost_admin.py         | 12      |
| GET /api/cost-admin/cache-summary                             | API route | app/api/cost_admin.py         | 12      |
| GET /api/cost-admin/infra-allocations                         | API route | app/api/cost_admin.py         | 12      |
| GET /api/cost-admin/overview                                  | API route | app/api/cost_admin.py         | 12      |
| GET /api/cost-admin/pricing                                   | API route | app/api/cost_admin.py         | 12      |
| GET /api/cost/run/{run_id}                                    | API route | app/api/cost.py               | 12      |
| GET /api/cost/session/{session_id}                            | API route | app/api/cost.py               | 12      |
| GET /api/entity-registry                                      | API route | app/api/entity_registry.py    | 12      |
| GET /api/entity-registry/entity-types                         | API route | app/api/entity_registry.py    | 12      |
| GET /api/eval/golden-set                                      | API route | app/api/eval.py               | 12      |
| GET /api/eval/history                                         | API route | app/api/eval.py               | 12      |
| GET /api/eval/workflow-golden-set                             | API route | app/api/eval.py               | 12      |
| GET /api/files                                                | API route | app/api/workflows.py          | 12      |
| GET /api/inspect/chunks                                       | API route | app/api/inspect.py            | 12      |
| GET /api/knowledge/collections                                | API route | app/api/knowledge.py          | 12      |
| GET /api/knowledge/collections/{collection_id}                | API route | app/api/knowledge.py          | 12      |
| GET /api/knowledge/collections/{collection_id}/documents      | API route | app/api/knowledge.py          | 12      |
| GET /api/knowledge/collections/{collection_id}/indexes        | API route | app/api/knowledge.py          | 12      |
| GET /api/knowledge/doc-types                                  | API route | app/api/knowledge.py          | 12      |
| GET /api/knowledge/documents/{document_id}                    | API route | app/api/knowledge.py          | 12      |
| GET /api/knowledge/documents/{document_id}/source-url         | API route | app/api/knowledge.py          | 12      |
|                                                               | API route | app/api/knowledge.py          | 12      |

| Entry                                                                      | Kind      | Primary source                              | Volumes |
|----------------------------------------------------------------------------|-----------|---------------------------------------------|---------|
| GET /api/knowledge/embedding-models                                        |           |                                             |         |
| GET /api/knowledge/ingestion-presets                                       | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| GET /api/knowledge/ingestions                                              | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| GET /api/knowledge/ingestions/{ingestion_job_id}                           | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| GET /api/knowledge/profiles                                                | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| GET /api/knowledge/profiles/{profile_id}                                   | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| GET /api/llm/models                                                        | API route | <a href="#">app/api/llm_providers.py</a>    | 12      |
| GET /api/llm/models/openrouter                                             | API route | <a href="#">app/api/llm_providers.py</a>    | 12      |
| GET /api/node-types                                                        | API route | <a href="#">app/api/workflows.py</a>        | 12      |
| GET /api/node-types/{node_type}/models                                     | API route | <a href="#">app/api/workflows.py</a>        | 12      |
| GET /api/prompt-templates                                                  | API route | <a href="#">app/api/prompt_templates.py</a> | 12      |
| GET /api/proposals/runs/{run_id}/review                                    | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| GET /api/proposals/{proposal_id}/approvals                                 | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| GET /api/proposals/{proposal_id}/sources/versions                          | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| GET /api/rag-agents                                                        | API route | <a href="#">app/api/rag_agents.py</a>       | 12      |
| GET /api/rag-agents/{rag_agent_id}                                         | API route | <a href="#">app/api/rag_agents.py</a>       | 12      |
| GET /api/research/skills                                                   | API route | <a href="#">app/api/research.py</a>         | 12      |
| GET /api/retrieval/presets                                                 | API route | <a href="#">app/api/retrieval.py</a>        | 12      |
| GET /api/retrieval/traces                                                  | API route | <a href="#">app/api/retrieval.py</a>        | 12      |
| GET /api/retrieval/traces/{retrieval_request_id}                           | API route | <a href="#">app/api/retrieval.py</a>        | 12      |
| GET /api/retrieval/traces/{retrieval_request_id}/chunks/{chunk_id}/context | API route | <a href="#">app/api/retrieval.py</a>        | 12      |
| GET /api/runs/mine                                                         | API route | <a href="#">app/api/runs.py</a>             | 12      |
| GET /api/runs/mine/{run_id}                                                | API route | <a href="#">app/api/runs.py</a>             | 12      |
| GET /api/runs/mine/{run_id}/chat                                           | API route | <a href="#">app/api/run_chat.py</a>         | 12      |
| GET /api/runs/mine/{run_id}/pending-gate                                   | API route | <a href="#">app/api/runs.py</a>             | 12      |
| GET /api/runs/{run_id}/events                                              | API route | <a href="#">app/api/workflows.py</a>        | 12      |
| GET /api/workflow-input-files/capabilities                                 | API route | <a href="#">app/api/workflow_files.py</a>   | 12      |
| GET /api/workflow-input-files/content                                      | API route | <a href="#">app/api/workflow_files.py</a>   | 12      |

| Entry                                           | Kind                   | Primary source                                                             | Volumes       |
|-------------------------------------------------|------------------------|----------------------------------------------------------------------------|---------------|
| GET /api/workflows                              | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/by-name/{name}               | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/chat-catalog                 | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/preflight-stats              | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/{name}/detail                | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/{name}/draft                 | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/{name}/stats                 | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/{name}/versions              | API route              | app/api/workflows.py                                                       | 12            |
| GET /api/workflows/{name}/versions/{version_id} | API route              | app/api/workflows.py                                                       | 12            |
| GET /auth/me                                    | API route              | app/api/auth.py                                                            | 12            |
| GET /health                                     | API route              | app/api/health.py                                                          | 12            |
| GET /ready                                      | API route              | app/api/health.py                                                          | 12            |
| GraphNormalizer                                 | Node type              | app/nodes/graph_normalizer.py                                              | 06, 17, 22    |
| HorizonDOCXProposalRenderer                     | Node type              | app/nodes/horizon_docx_renderer.py                                         | 06, 17, 22    |
| HorizonEvaluationAgent                          | Node type              | app/nodes/horizon_evaluation.py                                            | 06, 17, 22    |
| HorizonHTMLProposalRenderer                     | Node type              | app/nodes/html_proposal_renderer.py                                        | 06, 17, 22    |
| HumanInLoopAgent                                | Node type              | app/nodes/human_in_loop.py                                                 | 06, 17, 22    |
| IntegrationAgent                                | Node type              | app/nodes/integration.py                                                   | 06, 17, 22    |
| InternalProjectEvidenceRetrieverAgent           | Node type              | app/nodes/internal_project_evidence_retriever.py                           | 06, 17, 22    |
| KimiVisionAgent                                 | Node type              | app/nodes/kimi_vision_agent.py                                             | 06, 17, 22    |
| Knowledge Studio                                | Feature                | ui/src/modes/knowledge; app/api/knowledge.py; MinIO, Weaviate, collections | 22            |
| KnowledgeRetrieval                              | Node type              | app/nodes/knowledge_retrieval.py                                           | 06, 17, 22    |
| Literal                                         | Node type              | app/nodes/_stubs.py                                                        | 06, 17, 22    |
| LLM providers                                   | Feature                | model selectors; app/api/llm_providers.py; registry/router/cost            | 22            |
| MCP/integrations                                | Feature                | Builder tool panels; app/api/builder.py; MCP and integrations              | 22            |
| MCPAgent                                        | Node type              | app/nodes/mcp_agent.py                                                     | 06, 17, 22    |
| MCPToolAgent                                    | Node type              | app/nodes/mcp_tool.py                                                      | 06, 17, 22    |
| MethodologyEngineeringAgent                     | Node type              | app/nodes/methodology_engineering.py                                       | 06, 17, 22    |
| MinIOEvidenceIngestion                          | Node type              | app/nodes/minio_evidence_ingestion.py                                      | 06, 17, 22    |
| observability/grafana                           | Package/area (2 files) | observability/grafana                                                      | 01, 03/04, 17 |
| observability/prometheus                        | Package/area (1 files) | observability/prometheus                                                   | 01, 03/04, 17 |



| Entry                                                                   | Kind                   | Primary source                                                         | Volumes       |
|-------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------|---------------|
| observability/prometheus.yml                                            | Package/area (1 files) | observability/prometheus.yml                                           | 01, 03/04, 17 |
| OpenAllImageGenerationAgent                                             | Node type              | app/nodes/<br>openai_image_generation_agent.py                         | 06, 17, 22    |
| Operations                                                              | Feature                | health/run history UI; health/metrics<br>routes; deploy/compose/alerts | 22            |
| PaperQAEvidenceSynthesizerAgent                                         | Node type              | app/nodes/<br>paperqa_evidence_synthesizer.py                          | 06, 17, 22    |
| PATCH /api/builder/email/<br>connections/{connection_id}                | API route              | app/api/email_oauth.py                                                 | 12            |
| PDFProposalRenderer                                                     | Node type              | app/nodes/pdf_tool.py                                                  | 06, 17, 22    |
| PDFTextExtractor                                                        | Node type              | app/nodes/pdf_tool.py                                                  | 06, 17, 22    |
| POST /api/builder/assist/rules                                          | API route              | app/api/builder.py                                                     | 12            |
| POST /api/builder/assist/schema                                         | API route              | app/api/builder.py                                                     | 12            |
| POST /api/builder/mcp/test-tool                                         | API route              | app/api/builder.py                                                     | 12            |
| POST /api/builder/node-test                                             | API route              | app/api/builder.py                                                     | 12            |
| POST /api/builder/output-contract                                       | API route              | app/api/builder.py                                                     | 12            |
| POST /api/builder/schema-preview                                        | API route              | app/api/builder.py                                                     | 12            |
| POST /api/builder/simulate                                              | API route              | app/api/builder.py                                                     | 12            |
| POST /api/candidates/{run_id}/<br>verify-claim                          | API route              | app/api/candidates.py                                                  | 12            |
| POST /api/chat-conversations/<br>resolve                                | API route              | app/api/chat_conversations.py                                          | 12            |
| POST /api/chat-conversations/<br>{conversation_id}/messages             | API route              | app/api/chat_conversations.py                                          | 12            |
| POST /api/chat-workflows/from-<br>existing                              | API route              | app/api/chat_workflows.py                                              | 12            |
| POST /api/chat-workflows/generate                                       | API route              | app/api/chat_workflows.py                                              | 12            |
| POST /api/chat-workflows/import                                         | API route              | app/api/chat_workflows.py                                              | 12            |
| POST /api/chat-workflows/presets/<br>deep-research                      | API route              | app/api/chat_workflows.py                                              | 12            |
| POST /api/chat-workflows/presets/<br>general                            | API route              | app/api/chat_workflows.py                                              | 12            |
| POST /api/chat-workflows/<br>{chat_workflow_id}/request-<br>publication | API route              | app/api/chat_workflows.py                                              | 12            |
| POST /api/chat-workspace/plan                                           | API route              | app/api/chat_workspace.py                                              | 12            |
| POST /api/chat-workspace/prepare                                        | API route              | app/api/chat_workspace.py                                              | 12            |
| POST /api/chat-workspace/<br>workflow-labels                            | API route              | app/api/chat_workspace.py                                              | 12            |
| POST /api/entity-registry                                               | API route              | app/api/entity_registry.py                                             | 12            |
| POST /api/eval/run                                                      | API route              | app/api/eval.py                                                        | 12            |

| Entry                                                                       | Kind      | Primary source                              | Volumes |
|-----------------------------------------------------------------------------|-----------|---------------------------------------------|---------|
| POST /api/eval/score-output                                                 | API route | <a href="#">app/api/eval.py</a>             | 12      |
| POST /api/eval/workflow-compare                                             | API route | <a href="#">app/api/eval.py</a>             | 12      |
| POST /api/inspect/retrieve                                                  | API route | <a href="#">app/api/inspect.py</a>          | 12      |
| POST /api/knowledge/collections                                             | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/knowledge/collections/{collection_id}/indexes                     | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/knowledge/collections/{collection_id}/indexes/{index_id}/activate | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/knowledge/collections/{collection_id}/ingestions                  | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/knowledge/ingestions/{ingestion_job_id}/cancel                    | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/knowledge/profiles                                                | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/knowledge/profiles/defaults                                       | API route | <a href="#">app/api/knowledge.py</a>        | 12      |
| POST /api/llm/models/{model}/probe                                          | API route | <a href="#">app/api/llm_providers.py</a>    | 12      |
| POST /api/node-types/ask                                                    | API route | <a href="#">app/api/node_types_chat.py</a>  | 12      |
| POST /api/node-types/draft-code                                             | API route | <a href="#">app/api/node_types_chat.py</a>  | 12      |
| POST /api/node-types/draft-instructions                                     | API route | <a href="#">app/api/node_types_chat.py</a>  | 12      |
| POST /api/node-types/draft-prompt                                           | API route | <a href="#">app/api/node_types_chat.py</a>  | 12      |
| POST /api/prompt-templates                                                  | API route | <a href="#">app/api/prompt_templates.py</a> | 12      |
| POST /api/prompt-templates/{template_id}/duplicate                          | API route | <a href="#">app/api/prompt_templates.py</a> | 12      |
| POST /api/proposals/coverage                                                | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/approvals                                 | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/approvals/{approval_id}/decision          | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/concept-alternatives                      | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/evidence/verify                           | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/horizon-evaluation                        | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/render                                    | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/render/docx                               | API route | <a href="#">app/api/proposals.py</a>        | 12      |
| POST /api/proposals/{proposal_id}/sources/{source_id}/versions              | API route | <a href="#">app/api/proposals.py</a>        | 12      |

| Entry                                                               | Kind      | Primary source                                         | Volumes    |
|---------------------------------------------------------------------|-----------|--------------------------------------------------------|------------|
| POST /api/proposals/{proposal_id}/verify-claims                     | API route | <a href="#">app/api/proposals.py</a>                   | 12         |
| POST /api/rag-agents                                                | API route | <a href="#">app/api/rag_agents.py</a>                  | 12         |
| POST /api/rag-agents/{rag_agent_id}/query                           | API route | <a href="#">app/api/rag_agents.py</a>                  | 12         |
| POST /api/retrieval/compare                                         | API route | <a href="#">app/api/retrieval.py</a>                   | 12         |
| POST /api/retrieval/search                                          | API route | <a href="#">app/api/retrieval.py</a>                   | 12         |
| POST /api/runs/mine/{run_id}/chat                                   | API route | <a href="#">app/api/run_chat.py</a>                    | 12         |
| POST /api/runs/mine/{run_id}/pause                                  | API route | <a href="#">app/api/runs.py</a>                        | 12         |
| POST /api/runs/mine/{run_id}/resume                                 | API route | <a href="#">app/api/runs.py</a>                        | 12         |
| POST /api/runs/mine/{source_run_id}/restart                         | API route | <a href="#">app/api/runs.py</a>                        | 12         |
| POST /api/runs/mine/{source_run_id}/retry                           | API route | <a href="#">app/api/runs.py</a>                        | 12         |
| POST /api/workflow-input-files                                      | API route | <a href="#">app/api/workflow_files.py</a>              | 12         |
| POST /api/workflow-input-files/extract                              | API route | <a href="#">app/api/workflow_files.py</a>              | 12         |
| POST /api/workflows/autofix                                         | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/autofix-node                                    | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/generate                                        | API route | <a href="#">app/api/workflow_generation.py</a>         | 12         |
| POST /api/workflows/run                                             | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/save                                            | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/subprocess-callback/{token}                     | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/validate                                        | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/{name}/versions/{version_id}/restore            | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /api/workflows/{run_id}/resume                                 | API route | <a href="#">app/api/workflows.py</a>                   | 12         |
| POST /auth/logout                                                   | API route | <a href="#">app/api/auth.py</a>                        | 12         |
| POST /auth/token                                                    | API route | <a href="#">app/api/auth.py</a>                        | 12         |
| PowerPointProposalSlides                                            | Node type | <a href="#">app/nodes/powerpoint_tool.py</a>           | 06, 17, 22 |
| PriorProjectRetrieverAgent                                          | Node type | <a href="#">app/nodes/prior_project_retriever.py</a>   | 06, 17, 22 |
| ProposalEvidenceFactoryAgent                                        | Node type | <a href="#">app/nodes/proposal_evidence_factory.py</a> | 06, 17, 22 |
| ProposalSubmissionGate                                              | Node type | <a href="#">app/nodes/proposal_submission_gate.py</a>  | 06, 17, 22 |
| ProposalTruthGraphAgent                                             | Node type | <a href="#">app/nodes/proposal_truth_graph.py</a>      | 06, 17, 22 |
| PUT /api/chat-conversations/{conversation_id}/messages/{message_id} | API route | <a href="#">app/api/chat_conversations.py</a>          | 12         |

| Entry                                            | Kind                   | Primary source                                                  | Volumes       |
|--------------------------------------------------|------------------------|-----------------------------------------------------------------|---------------|
| PUT /api/cost-admin/budgets/global               | API route              | app/api/cost_admin.py                                           | 12            |
| PUT /api/cost-admin/budgets/session/{session_id} | API route              | app/api/cost_admin.py                                           | 12            |
| PUT /api/cost-admin/infra-allocations/{model}    | API route              | app/api/cost_admin.py                                           | 12            |
| PUT /api/cost-admin/pricing/{model}              | API route              | app/api/cost_admin.py                                           | 12            |
| PUT /api/prompt-templates/{template_id}          | API route              | app/api/prompt_templates.py                                     | 12            |
| PUT /api/prompt-templates/{template_id}/favorite | API route              | app/api/prompt_templates.py                                     | 12            |
| PUT /api/workflows/{name}/draft                  | API route              | app/api/workflows.py                                            | 12            |
| pyproject.toml                                   | Package/area (1 files) | pyproject.toml                                                  | 01, 03/04, 17 |
| PythonSnippetAgent                               | Node type              | app/nodes/python_snippet.py                                     | 06, 17, 22    |
| RAG agents                                       | Feature                | retrieval selectors; app/api/rag_agents.py; RAG/retrieval/nodes | 22            |
| RAGAgent                                         | Node type              | app/nodes/rag.py                                                | 06, 17, 22    |
| README.md                                        | Package/area (1 files) | README.md                                                       | 01, 03/04, 17 |
| ResearchSourceAcquirer                           | Node type              | app/nodes/research_source_acquirer.py                           | 06, 17, 22    |
| RouterAgent                                      | Node type              | app/nodes/router.py                                             | 06, 17, 22    |
| ScientificResearchPlannerAgent                   | Node type              | app/nodes/scientific_research_planner.py                        | 06, 17, 22    |
| ScientificSkillAgent                             | Node type              | app/nodes/scientific_skill_agent.py                             | 06, 17, 22    |
| scripts/backfill_knowledge_resources.py          | Package/area (1 files) | scripts/backfill_knowledge_resources.py                         | 01, 03/04, 17 |
| scripts/build_documentation_pdf.py               | Package/area (1 files) | scripts/build_documentation_pdf.py                              | 01, 03/04, 17 |
| scripts/build_eu_proposal_copy_bundle.py         | Package/area (1 files) | scripts/build_eu_proposal_copy_bundle.py                        | 01, 03/04, 17 |
| scripts/build_hitl_editor_bundle.py              | Package/area (1 files) | scripts/build_hitl_editor_bundle.py                             | 01, 03/04, 17 |
| scripts/build_workflow_file_inputs_bundle.py     | Package/area (1 files) | scripts/build_workflow_file_inputs_bundle.py                    | 01, 03/04, 17 |
| scripts/build_workflow_qa_matrix.py              | Package/area (1 files) | scripts/build_workflow_qa_matrix.py                             | 01, 03/04, 17 |
| scripts/check_api_routes.py                      | Package/area (1 files) | scripts/check_api_routes.py                                     | 01, 03/04, 17 |
| scripts/debug_compressor.py                      | Package/area (1 files) | scripts/debug_compressor.py                                     | 01, 03/04, 17 |
| scripts/generate_docstrings.py                   | Package/area (1 files) | scripts/generate_docstrings.py                                  | 01, 03/04, 17 |
| scripts/generate_fixtures.py                     | Package/area (1 files) | scripts/generate_fixtures.py                                    | 01, 03/04, 17 |
| scripts/generate_production_env.py               | Package/area (1 files) | scripts/generate_production_env.py                              | 01, 03/04, 17 |
| scripts/generate_reference_workflows.py          | Package/area (1 files) | scripts/generate_reference_workflows.py                         | 01, 03/04, 17 |

| Entry                                      | Kind                   | Primary source                                                 | Volumes       |
|--------------------------------------------|------------------------|----------------------------------------------------------------|---------------|
| scripts/ingest_samples.py                  | Package/area (1 files) | <a href="#">scripts/ingest_samples.py</a>                      | 01, 03/04, 17 |
| scripts/inspect_weaviate.py                | Package/area (1 files) | <a href="#">scripts/inspect_weaviate.py</a>                    | 01, 03/04, 17 |
| scripts/live_provider_smoke.py             | Package/area (1 files) | <a href="#">scripts/live_provider_smoke.py</a>                 | 01, 03/04, 17 |
| scripts/load_test.py                       | Package/area (1 files) | <a href="#">scripts/load_test.py</a>                           | 01, 03/04, 17 |
| scripts/manage_user.py                     | Package/area (1 files) | <a href="#">scripts/manage_user.py</a>                         | 01, 03/04, 17 |
| scripts/new_node_type.py                   | Package/area (1 files) | <a href="#">scripts/new_node_type.py</a>                       | 01, 03/04, 17 |
| scripts/openrouter_health_check.py         | Package/area (1 files) | <a href="#">scripts/openrouter_health_check.py</a>             | 01, 03/04, 17 |
| scripts/preflight_workflows.py             | Package/area (1 files) | <a href="#">scripts/preflight_workflows.py</a>                 | 01, 03/04, 17 |
| scripts/production_preflight.py            | Package/area (1 files) | <a href="#">scripts/production_preflight.py</a>                | 01, 03/04, 17 |
| scripts/register_biomass.py                | Package/area (1 files) | <a href="#">scripts/register_biomass.py</a>                    | 01, 03/04, 17 |
| scripts/render_sample_proposal.py          | Package/area (1 files) | <a href="#">scripts/render_sample_proposal.py</a>              | 01, 03/04, 17 |
| scripts/<br>render_sample_proposal_docx.py | Package/area (1 files) | <a href="#">scripts/render_sample_proposal_docx.py</a>         | 01, 03/04, 17 |
| scripts/retag_work_programme.py            | Package/area (1 files) | <a href="#">scripts/retag_work_programme.py</a>                | 01, 03/04, 17 |
| scripts/seed_collections.py                | Package/area (1 files) | <a href="#">scripts/seed_collections.py</a>                    | 01, 03/04, 17 |
| scripts/smoke_events.py                    | Package/area (1 files) | <a href="#">scripts/smoke_events.py</a>                        | 01, 03/04, 17 |
| scripts/smoke_llm.py                       | Package/area (1 files) | <a href="#">scripts/smoke_llm.py</a>                           | 01, 03/04, 17 |
| scripts/smoke_production.py                | Package/area (1 files) | <a href="#">scripts/smoke_production.py</a>                    | 01, 03/04, 17 |
| scripts/smoke_retrieval.py                 | Package/area (1 files) | <a href="#">scripts/smoke_retrieval.py</a>                     | 01, 03/04, 17 |
| scripts/smoke_structured.py                | Package/area (1 files) | <a href="#">scripts/smoke_structured.py</a>                    | 01, 03/04, 17 |
| scripts/<br>start_snippet_runner_local.sh  | Package/area (1 files) | <a href="#">scripts/start_snippet_runner_local.sh</a>          | 01, 03/04, 17 |
| scripts/<br>verder_knowledge_config.json   | Package/area (1 files) | <a href="#">scripts/verder_knowledge_config.json</a>           | 01, 03/04, 17 |
| scripts/verder_process_emails.py           | Package/area (1 files) | <a href="#">scripts/verder_process_emails.py</a>               | 01, 03/04, 17 |
| scripts/<br>verder_reingest_production.py  | Package/area (1 files) | <a href="#">scripts/verder_reingest_production.py</a>          | 01, 03/04, 17 |
| scripts/verder_setup_knowledge.py          | Package/area (1 files) | <a href="#">scripts/verder_setup_knowledge.py</a>              | 01, 03/04, 17 |
| scripts/verify_knowledge_studio.py         | Package/area (1 files) | <a href="#">scripts/verify_knowledge_studio.py</a>             | 01, 03/04, 17 |
| SQLQueryAgent                              | Node type              | <a href="#">app/nodes/sql_query.py</a>                         | 06, 17, 22    |
| StartAgent                                 | Node type              | <a href="#">app/nodes/start.py</a>                             | 06, 17, 22    |
| StructuredDatasetRetrieverAgent            | Node type              | <a href="#">app/nodes/<br/>structured_dataset_retriever.py</a> | 06, 17, 22    |
| SubprocessAgent                            | Node type              | <a href="#">app/nodes/subprocess.py</a>                        | 06, 17, 22    |
| TextAssemblerAgent                         | Node type              | <a href="#">app/nodes/text_assembler.py</a>                    | 06, 17, 22    |
| TransformAgent                             | Node type              | <a href="#">app/nodes/transform.py</a>                         | 06, 17, 22    |
| ui/e2e                                     | Package/area (4 files) | <a href="#">ui/e2e</a>                                         | 01, 03/04, 17 |

| Entry                      | Kind                     | Primary source                                                                                 | Volumes       |
|----------------------------|--------------------------|------------------------------------------------------------------------------------------------|---------------|
| ui/src                     | Package/area (256 files) | ui/src                                                                                         | 01, 03/04, 17 |
| Visual workflow Builder    | Feature                  | ui/src/modes/studio/Builder.tsx; app/api/builder.py; workflows.py; workflow files and versions | 22            |
| WebSearchAgent             | Node type                | app/nodes/web_search_agent.py                                                                  | 06, 17, 22    |
| Workflow execution/Cockpit | Feature                  | ui/src/modes/studio/Cockpit.tsx; app/api/runs.py; runtime/events/run history                   | 22            |
| WorkflowFileLoader         | Node type                | app/nodes/workflow_file_loader.py                                                              | 06, 17, 22    |
| WorkflowInputAgent         | Node type                | app/nodes/workflow_input.py                                                                    | 06, 17, 22    |